

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460923.7219 +/- 0.0069 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.045 +/- 0.062
Transit depth (Rp/Rs)^2	0.2 +/- 0.55 [%]
Orbital Inclination (inc)	84.92 +/- 1.9
Airmass coefficient 1 (a1)	0.182 +/- 0.091
Airmass coefficient 2 (a2)	1.43 +/- 0.58
Scatter in the residuals of the lightcurve fit is	51.32 %
Best Comparison Star	#2 - [255, 198]
Optimal Aperture	2.53
Optimal Annulus	7.0
Transit Duration (day)	0.031 +/- 0.032

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.045 ± 0.062 vs 0.1592 (deviation 1.84σ)
Duration fit vs published	0.031 d vs 0.0946 d (deviation 1.99σ)
Mid-time O–C	3.6 min
Depth SNR	0.7
Residual scatter	51.32 %

Light curve

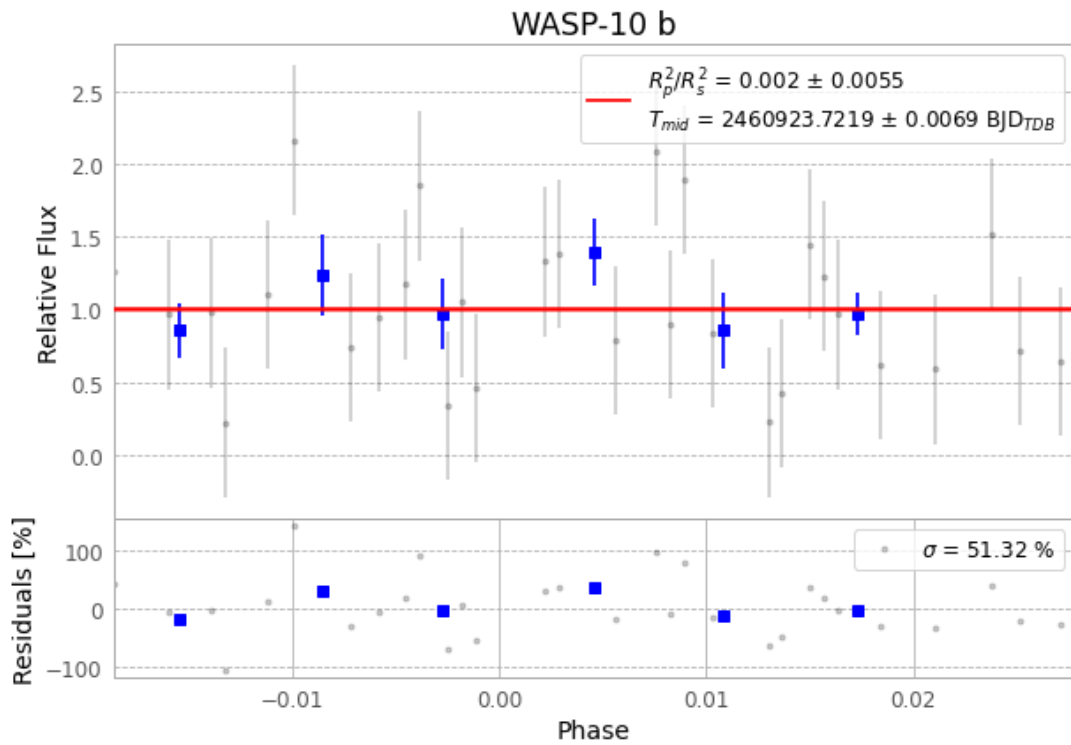


Figure 1 — FinalLightCurve_WASP-10 b_2025-09-04.png (SHA-256 ba50173bcfd96633...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [555, 322], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 87895b02142a6f4c...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

72 FITS frames (47 MB), WASP-10250905034516.FITS → WASP-10250905071816.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-10-250905.log (1317bfb531dc...), FinalLightCurve_WASP-10 b_2025-09-04.png (ba50173bcfd9...), FinalLightCurve_WASP-10 b_2025-09-04.csv (e2fe8b729a9f...)

Compute resources

Wall time 130.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 18:20Z.